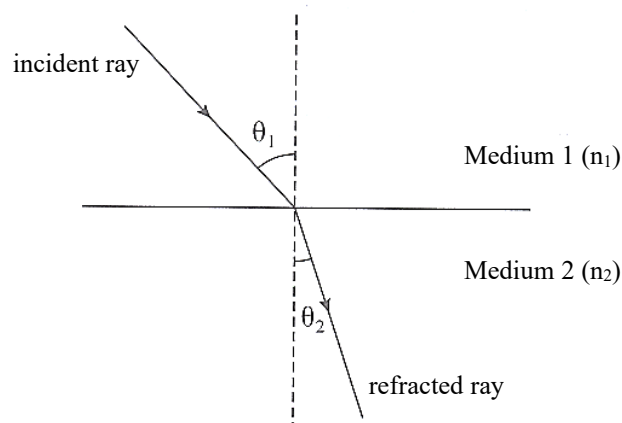


Refraction of light

- Light travels with the greatest speed in vacuum or air.
- **Refraction** occurs when light travels from one medium to another medium with **different speed**.
- During refraction, frequency of light f remains unchanged.
- When refraction occurs, the light ray will change its direction except for normal incidence.
- Refractive index (n) of a material (other than air) must be greater than 1. A medium of smaller refractive index is called an **optically less dense** medium but a medium of larger refractive index is called **optically denser medium**.

Snell's law:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$



Law of refraction:

- The normal, incident ray and refracted ray all lie on the same plane.

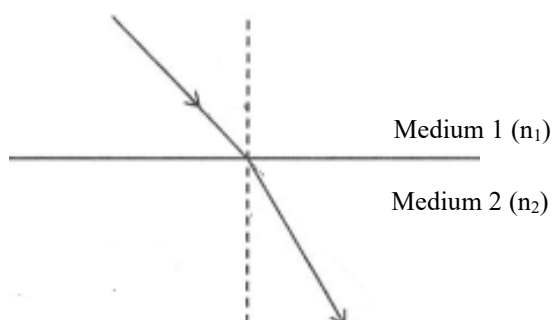
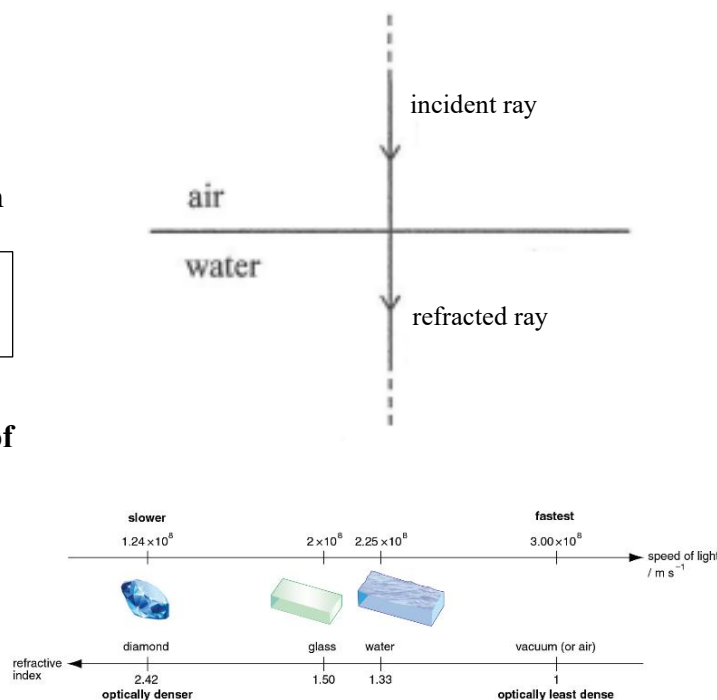
$$\frac{\sin i}{\sin r} = \text{constant}$$

i = angle of incidence
 r = angle of refraction

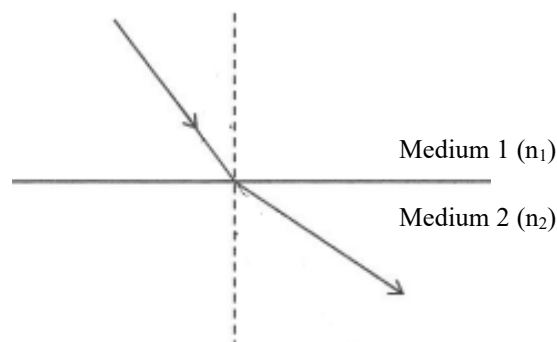
- The refractive index can be expressed in terms of speed of light:

$$n = \frac{\text{speed of light in a vacuum } (c)}{\text{speed of light in the medium } (v)}$$

- The **greater** the **refractive index** of a medium, the **lower** is the **speed** of light in the medium.



The refracted ray **bend towards normal** when the ray travels from **optically less dense** medium into **optically denser** medium.



The refracted ray **bend away from normal** when the ray travels from **optically denser** medium into **optically less dense** medium.

1. Which of the following statements about the refraction of light as shown in the sketch below is/are correct?

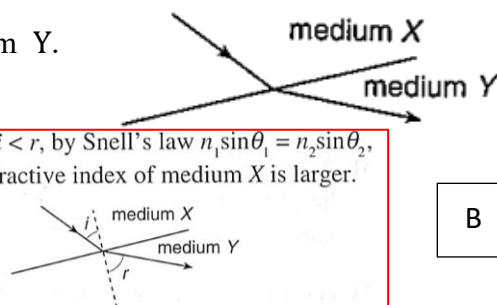
- (1) The angle of incidence is larger than the angle of refraction.
- (2) Light travels faster in medium X than in medium Y.
- (3) The refractive index of medium X is greater than that of medium Y.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

(1): Both the angle of incidence and the angle of refraction are measured from the normal which is perpendicular to the interface as shown.

(2): The light ray is deflected away from the normal. This indicates that light travels faster in the second medium.

(3): Since $i < r$, by Snell's law $n_1 \sin \theta_1 = n_2 \sin \theta_2$, the refractive index of medium X is larger.



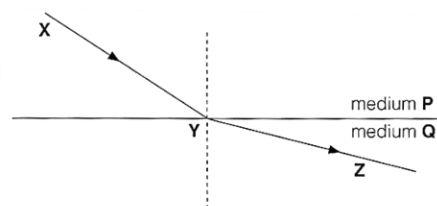
B

2. A ray of light XY travels from medium P into medium Q as shown below.

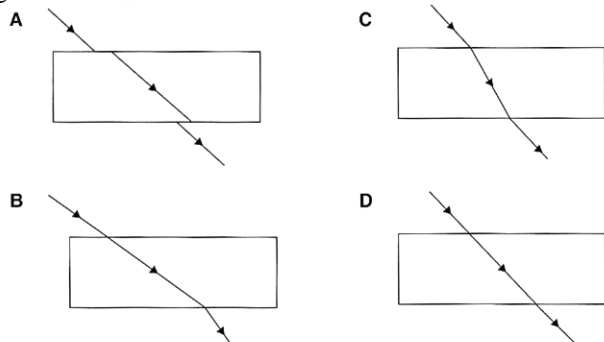
Which of the following statement is correct?

- A. Medium P is less dense than medium Q and the ray YZ travels faster than XY.
- B. Medium P is denser than medium Q and the ray YZ travels faster than XY.
- C. Medium P is less dense than medium Q and the ray YZ travels slower than XY.
- D. Medium P is denser than medium Q and the ray YZ travels slower than XY.

B



3. Which of the following correctly shows the path taken by a beam of light as it passes through a block of glass?



C

4. A ray of light passes from air to glass. The angle of incidence is 30° . The refractive index of the glass is 1.5. What is the angle of refraction?

- A. 19.5°
- B. 20°
- C. 45°
- D. 48.6°

$$\text{By } n = \frac{\sin \theta_a}{\sin \theta}, \quad \sin \theta = \frac{\sin \theta_a}{n} = \frac{\sin 30^\circ}{1.5}$$

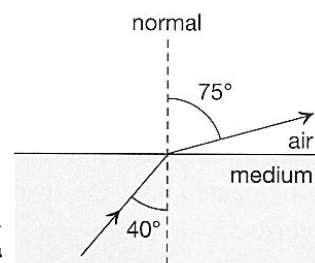
$$\theta = 19.5^\circ$$

A

5. The figure shows a light ray travelling from a medium to air. Find the refractive index of the medium.

- A. 0.67
- B. 0.70
- C. 1.43
- D. 1.50

$$\frac{\sin 40^\circ}{\sin 75^\circ} = \frac{1}{n}$$



D

6. A light ray emerges from a liquid surface into air with an angle of refraction of 60° . If the refractive index of the unknown liquid is 1.15, what is the angle of incidence?

- A. 20.2°
B. 30.0°
C. 35.1°
D. 48.9°

The angle of refraction is the angle in air,

$$\therefore \theta_{\text{air}} = 60^\circ.$$

$$n_{\text{liquid}} = \frac{\sin \theta_{\text{air}}}{\sin \theta_{\text{liquid}}}$$

$$1.15 = \frac{\sin 60^\circ}{\sin \theta_{\text{liquid}}}$$

$$\theta_{\text{liquid}} = 48.9^\circ$$

D

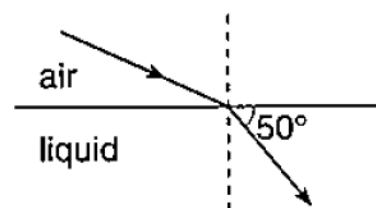
7. The given ray diagram shows the refraction of light at the surface of an unknown liquid. If the refractive index of the liquid is 1.42, what is the angle of incidence?

- A. 26.9°
B. 32.6°
C. 40°
D. 65.9°

$$n_{\text{liquid}} = \frac{\sin \theta_{\text{air}}}{\sin \theta_{\text{liquid}}}$$

$$1.42 = \frac{\sin \theta_{\text{air}}}{\sin (90^\circ - 50^\circ)}$$

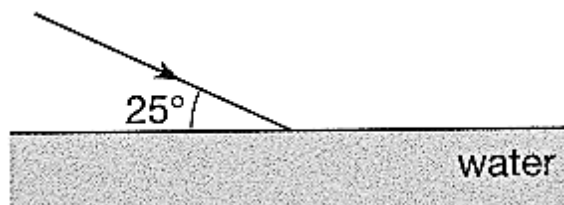
$$\theta_{\text{air}} = 65.9^\circ$$



D

8. A light ray strikes a water surface at an angle of 25° as shown above. By what angle will the ray be deflected after passing through the water surface? (Given : Refractive index of water = 1.33)

- A. 18.5° towards the normal
B. 22.0° towards the normal
C. 43.0° towards the normal
D. 47.0° towards the normal



B

Snell's law:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\sin (90^\circ - 25^\circ) = 1.33 \sin \theta_2$$

$$\sin \theta_2 = 0.6814$$

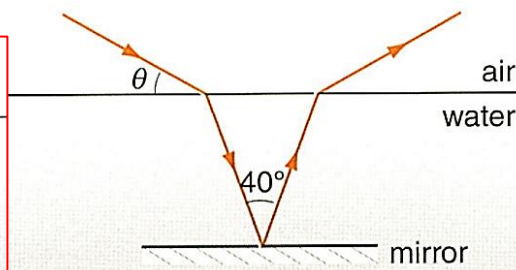
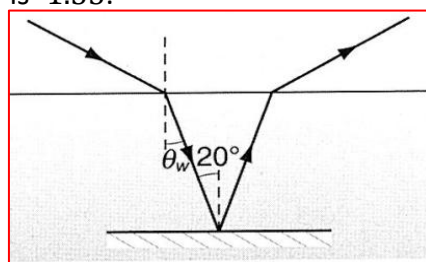
$$\theta_2 = 42.96^\circ$$

The ray is deflected by $65^\circ - 42.96^\circ = 22.0^\circ$.

9. A light ray travels from air into water and is reflected by a horizontal mirror in the water as shown. The refractive index of water is 1.33.

What is the value of θ ?

- A. 14.9°
B. 27.1°
C. 62.9°
D. 75.1°



C

Consider the ray reflected by the mirror. By the laws of reflection,

$$\text{angle of incidence} = \text{angle of reflection} = \frac{40^\circ}{2} = 20^\circ$$

Consider the air-water interface.

Angle between ray and normal in water = 20° (alt $\angle$ s, // lines)

$$\text{By } n = \frac{\sin \theta_a}{\sin \theta_w},$$

$$\sin \theta_a = n \sin \theta_w$$

$$= 1.33 \times \sin 20^\circ$$

$$\theta_a = 27.1^\circ$$

$$\theta = 90^\circ - 27.1^\circ = 62.9^\circ$$

10. The diagram above shows the path of a light ray travelling through media X, Y and Z. Arrange the media in the ascending order of refractive index.

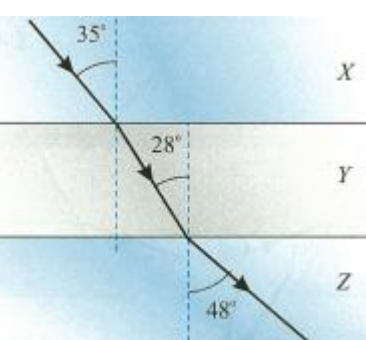
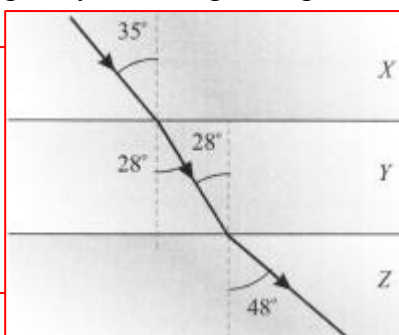
- A. X, Y, Z
B. Y, X, Z
C. Z, Y, X
D. Z, X, Y

Consider refraction at each interface:

$$\begin{cases} n_X \sin 35^\circ = n_Y \sin 28^\circ & \dots\dots(1) \\ n_Y \sin 28^\circ = n_Z \sin 48^\circ & \dots\dots(2) \end{cases}$$

$$\Rightarrow n_X = 0.818n_Y = 1.296n_Z$$

$$\therefore n_Z < n_X < n_Y$$



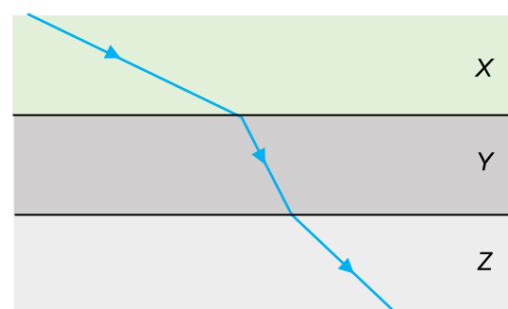
D

11. A ray of light passes through media X, Y and Z as shown.

Which of the following statements about the three media are correct?

- (1) Z is optically denser than X.
(2) Light travels the fastest in X.
(3) Y has the greatest refractive index.

- A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)



D

12. A light ray passes through a spherical air bubble in water. Which of the following represents the path of the emergent ray?

- A. P
B. Q
C. R
D. S

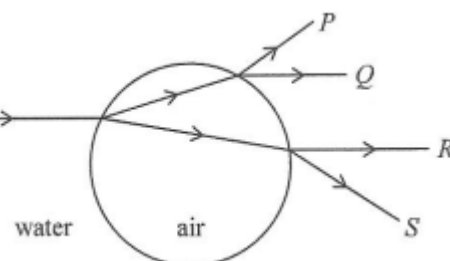
Normal is the line passing through the contact point and the centre of sphere.

From water to air: denser medium to less dense medium

$\Rightarrow$ bend away from normal $\Rightarrow$ the upper light ray is correct

From air to water: less dense medium to denser medium

$\Rightarrow$ bend towards normal $\Rightarrow$ ray P is correct



A

13. When a light ray travels from air to water with an angle of incidence of 45° , the angle of refraction is 32.1° . What is the speed of light in the water?

- A. $2.14 \times 10^8 \text{ m s}^{-1}$
B. $2.25 \times 10^8 \text{ m s}^{-1}$
C. $2.42 \times 10^8 \text{ m s}^{-1}$
D. $2.59 \times 10^8 \text{ m s}^{-1}$

$$\text{By } n = \frac{\sin \theta_a}{\sin \theta} = \frac{c}{v},$$

$$v = c \times \frac{\sin \theta}{\sin \theta_a}$$

$$= 3 \times 10^8 \times \frac{\sin 32.1^\circ}{\sin 45^\circ}$$

$$= 2.25 \times 10^8 \text{ m s}^{-1}$$

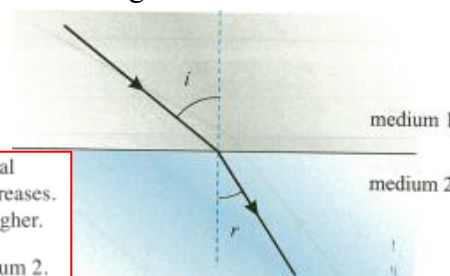
B

14. A beam of light travels across two media as shown above. Which of the following statements is/are correct?

- (1) The refractive index of medium 1 is greater than that of medium 2
(2) The speed of light decreases as it travels across the interface
(3) If i increases, r will also increase

- A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

- (1) is incorrect. The ray bends towards the normal during refraction. It means that its speed decreases. The refractive index of medium 2 must be higher.
(2) is correct. The wavelength is shorter in medium 2. The speed of light is lower in medium 2.
(3) is correct. According to $n = \frac{\sin i}{\sin r}$, as i increases, r will also increase.



D

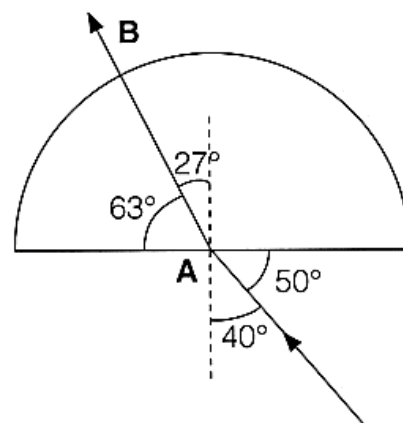
15. The diagram below shows a ray of red light being refracted at points A and B on a semi-circular glass block.

- What is the angle of incidence at A?
- What is the angle of refraction at A?
- Why was there no change in direction at B?

(b) 40°

(c) 27°

(a) The incident ray is on the normal. The angle of incidence is zero, so the angle of refraction is also zero.



16. The diagram below shows a ray of light incident on a boundary between two mediums.

- Draw arrows on the two light rays to show the directions in which the light rays would travel.
- Draw the normal at P and label it N.
- State the size of the following angles:
 - Angle of incidence

(ii) Angle of reflection

(iii) Angle of refraction

(c) (i) 35°
(ii) 35°
(iii) 22°

- If the two mediums are air and glass, which of the mediums X or Y, is the glass?

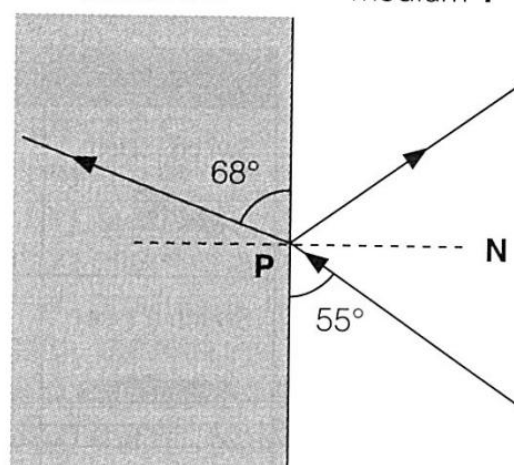
Explain your answer.

(d) Medium X. Light bends towards the normal as it travels into a denser medium.

(a), (b)

medium X

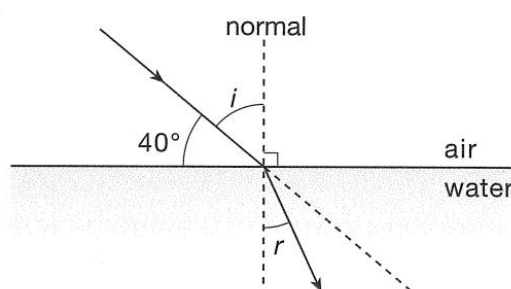
medium Y



17. A narrow ray of light strikes a smooth water surface. The ray makes an angle of 40° with the water surface. Given that the refractive index of water is 1.33.

- Draw a ray diagram to show the refraction at the air-water boundary.
- Find the angle of incident and angle of refraction.

(a) The incident ray makes an angle of 40° with the boundary.
It is bent towards the normal when it travels from air to water.



(b)

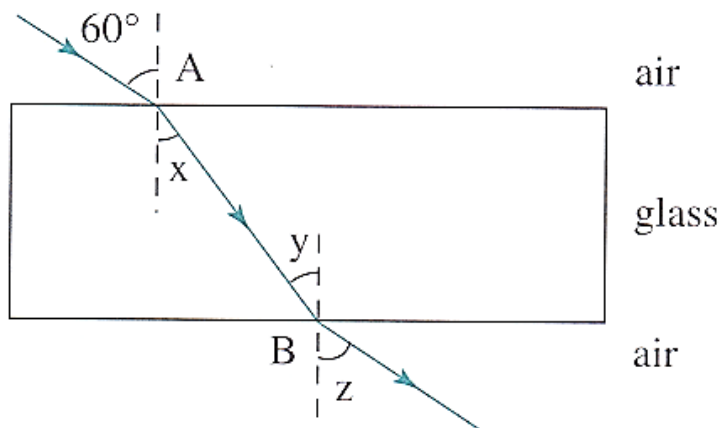
$$i = 90^\circ - 40^\circ = 50^\circ$$

$$n = \frac{\sin i}{\sin r}$$

$$\sin r = \frac{\sin 50^\circ}{1.33}$$

$$r = 35.2^\circ$$

18. Find the angles x, y and z .
(Refractive index of glass = 1.5)
(Refractive index of air = 1)



$$\begin{aligned} x &= 35.3^\circ \\ y &= 35.3^\circ \\ z &= 60^\circ \end{aligned}$$

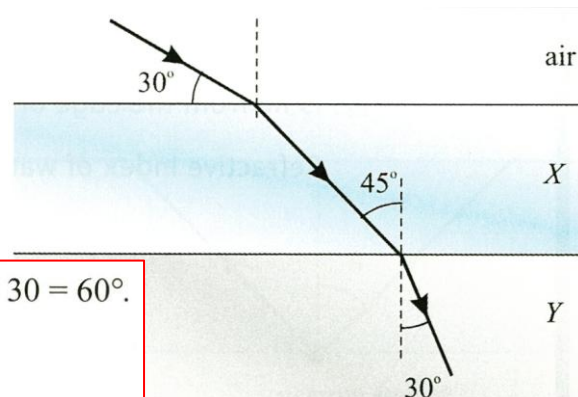
19. The speed of light in a medium is $2 \times 10^8 \text{ m s}^{-1}$.
(a) Find the refractive index of the medium.
(b) A light ray travels from the medium to air. The angle of refraction is 45° .
Find the angle of incidence.

$$\begin{aligned} \text{(a) Refractive index} &= \frac{c}{v} \\ &= \frac{3 \times 10^8}{2 \times 10^8} \\ &= 1.5 \end{aligned}$$

$$\begin{aligned} \text{(b) By } n &= \frac{\sin \theta_a}{\sin \theta}, \\ \sin \theta &= \frac{\sin \theta_a}{n} = \frac{\sin 45^\circ}{1.5} \\ \theta &= 28.1^\circ \\ \therefore \text{ The angle of incidence is } 28.1^\circ. \end{aligned}$$

20. A ray of light travels in air and passes through two media, X and Y , as shown.

- (a) Find the refractive index of X .
(b) Find the ratio $\frac{\text{Speed of light in } X}{\text{Speed of light in } Y}$.



- (a) At the air- X interface, the angle of incidence is $i = 90 - 30 = 60^\circ$.

The angle of refraction is $r = 45^\circ$.

$$\text{Refractive index of } X \text{ is } n_x = \frac{\sin i}{\sin r} = \frac{\sin 60^\circ}{\sin 45^\circ} = \underline{\underline{1.22}}$$

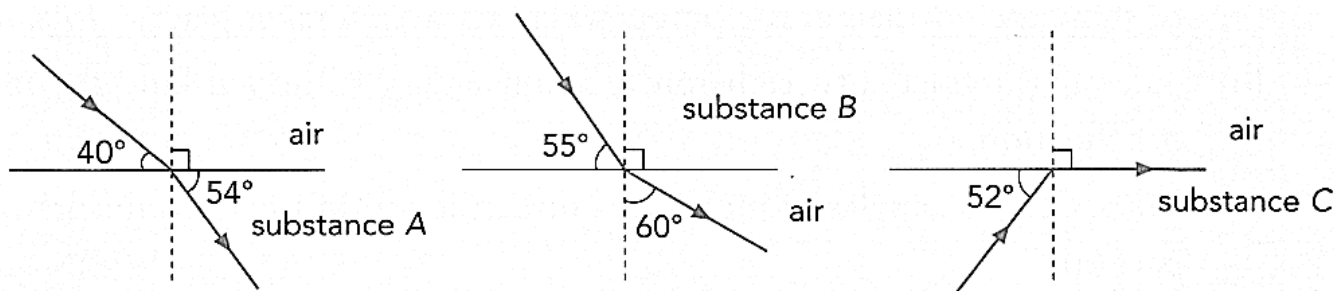
- (b) By $n_1 \sin \theta_1 = n_2 \sin \theta_2$, the refractive index of Y is given by

$$n_y \sin 30^\circ = 1.22 \times \sin 45^\circ$$

$$\therefore n_y = 1.73$$

$$\text{Since } v \propto \frac{1}{n}, \frac{\text{speed of light in } X}{\text{speed of light in } Y} = \frac{n_y}{n_x} = \frac{1.73}{1.22} = \underline{\underline{1.42}}$$

21. (a) In each of the following cases, a light ray passes from air into a substance or vice versa. Find the refractive indices of the media.



- (b) The three substances in part (a) are arranged to form parallel boundaries as shown in the following diagram. A ray of light is incident on substance A.

The ray emerges from substance C as shown.

- (i) Draw and complete the path of ray in the substances.
(ii) What is the angle of incidence when the ray enters substance A?

(a) Refractive index $\eta = \frac{\sin i}{\sin r}$

where i = angle between the ray and the normal in air

r = angle between the ray and the normal in medium

Refractive index of substance A,

$$\eta_A = \frac{\sin (90^\circ - 40^\circ)}{\sin (90^\circ - 54^\circ)} \quad (1)$$

$$= 1.30 \quad (1)$$

Refractive index of substance B,

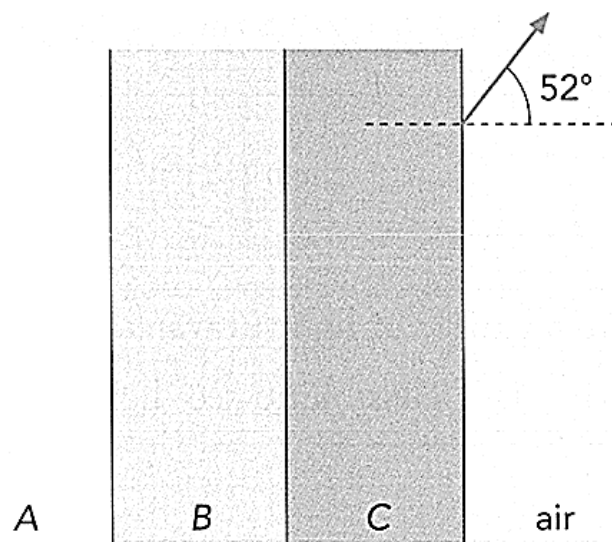
$$\eta_B = \frac{\sin 60^\circ}{\sin (90^\circ - 55^\circ)}$$

$$= 1.51$$

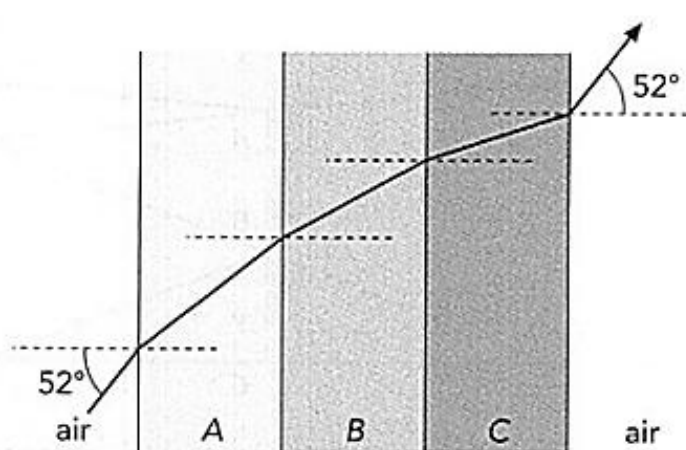
Refractive index of substance C,

$$\eta_C = \frac{\sin 90^\circ}{\sin (90^\circ - 52^\circ)}$$

$$= 1.62$$

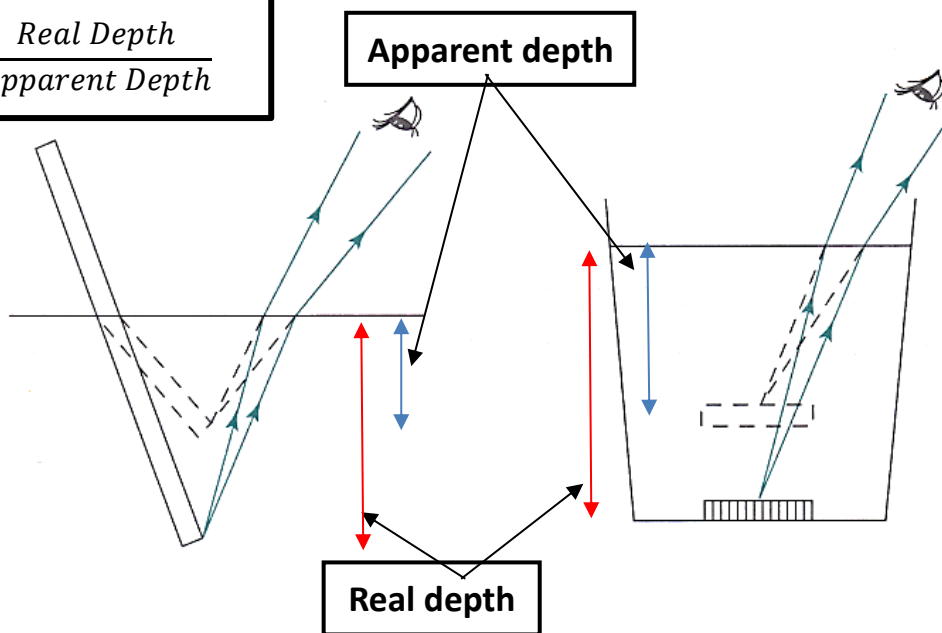


(b) (i)

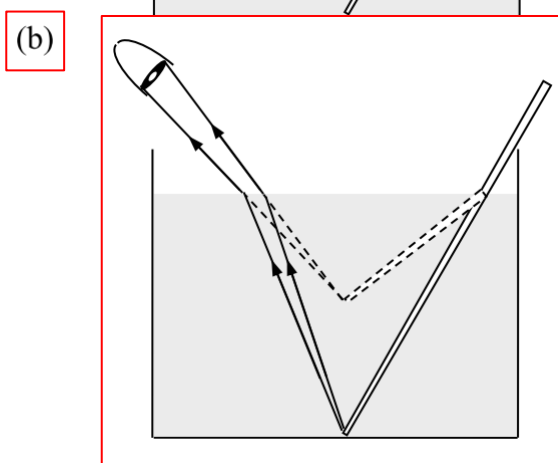
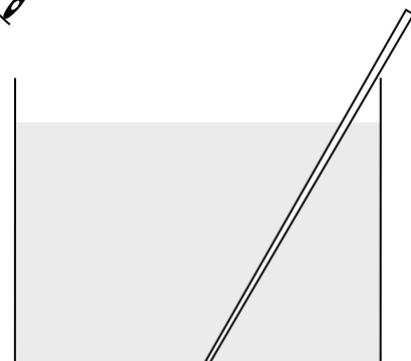
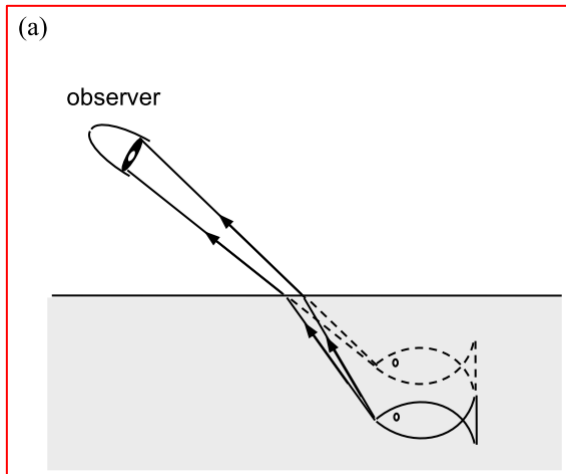
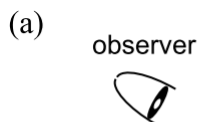


(ii) The angle of incidence is 52° .

$$\text{Refractive index } (n) = \frac{\text{Real Depth}}{\text{Apparent Depth}}$$



22. Draw a ray diagram in each of the following figures to show how the observer sees the object.



23. A boy is looking at the small print on a piece of paper. Meanwhile, he moves a glass block of uniform thickness 3 cm over the print. Then the print appears to rise by 0.9 cm.

(a) Draw a ray diagram to show how the boy views the print through the glass block.

(b) What is the nature of the image?

(c) Calculate the refractive index of glass.

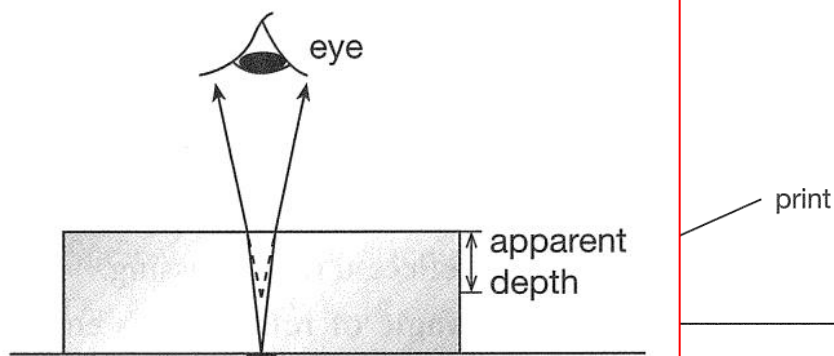
(b) Virtual and erect

$$(c) \quad n = \frac{\text{Real depth}}{\text{Apparent depth}}$$

$$= \frac{3}{3 - 0.9}$$

$$= 1.43$$

(a)

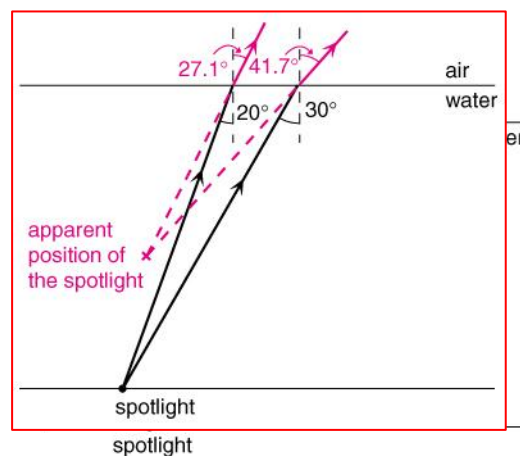


24. In the figure below, the two rays represent the cone of rays coming from the spotlight which is fixed at the bottom of the pool.

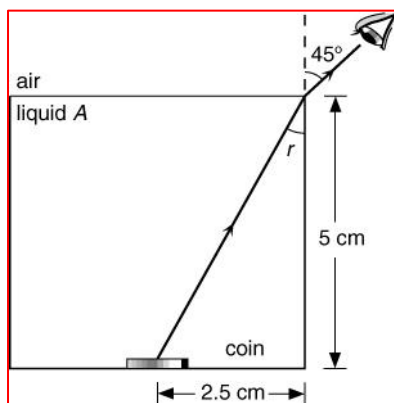
(a) Complete the paths of the two light rays in the air.

(b) Use the answer in part (a) to explain why the pool appears to be shallower than it actually is.

(b) As observed from the answer in part (a), when the light travel from water to air, it bends away from the normal (1A) Spot light appears to be nearer to observer and come from a position higher than the bottom of the pool (1A). This explains why the pool appears to be shallower than it actually is.



25. The length of each side of an opaque cubic container is 5 cm and a coin is placed at the centre of the base as shown. At the instant shown, the observer is unable to see the coin. However, the coin can just be seen when the container is fully filled with liquid A. Find the refractive index of liquid A.



$$\tan r = \frac{2.5}{5}$$

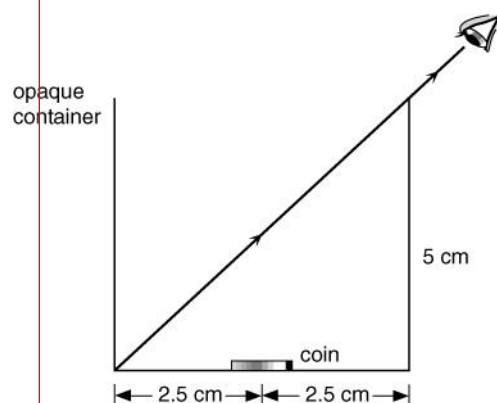
$$r = 26.57^\circ$$

From the definition of refractive index,

$$n = \frac{\sin i}{\sin r}$$

$$n = \frac{\sin 45^\circ}{\sin 26.57^\circ}$$

$$n = 1.581$$



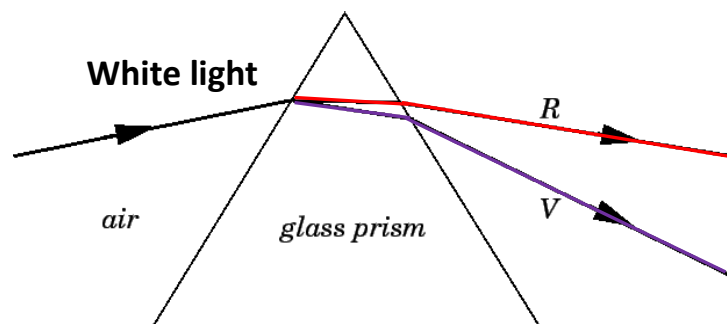
9 The refractive index n of liquid A is 1.58.

警告：不可翻印，販賣，違者必究

Dispersion of light

White light passing through a prism is refracted and separated into different colour. Different colour travel with different speed in the prism.

- Violet light bends more than red light
- Refractive index (n) for violet is the largest and for red is the smallest
- Speed of violet is the lowest and speed of red is the highest



26. Arrange the refractive indices of glass for orange, green and blue light in descending order.

- n_B : Refractive index for blue light
 n_G : Refractive index for green light
 n_O : Refractive index for orange light

- A. n_B, n_G, n_O
 B. n_B, n_O, n_G
 C. n_G, n_B, n_O
 D. n_G, n_O, n_B

A

27. Arrange in ascending order the speed of yellow, violet and blue light in glass.

- V_Y : Speed of yellow light
 V_V : Speed of violet light
 V_B : Speed of blue light

- A. V_Y, V_V, V_B
 B. V_Y, V_B, V_V
 C. V_V, V_Y, V_B
 D. V_V, V_B, V_Y

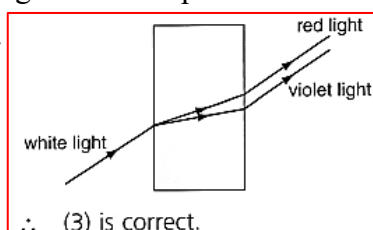
D

28. A beam of white light is separated into different colours after passing through a prism.

Which of the following statements is/are correct?

- (1) The refractive index of the prism for red light is smaller than that for violet light.
 (2) Violet light travels with the same speed as red light in the prism.
 (3) A beam of white light can be separated into different colours if it enters a rectangular glass block at an angle.

- A. (1) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. All of the above.



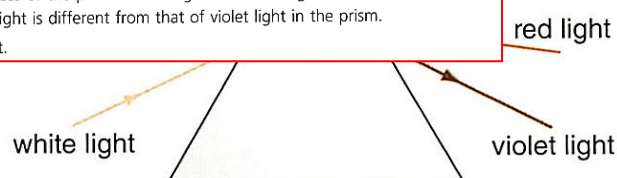
Red light is refracted less than violet light.

$\therefore$ (1) is correct.

The refractive indices of the prism for red light and violet light are different. Therefore, the speed of red light is different from that of violet light in the prism.

$\therefore$ (2) is incorrect.

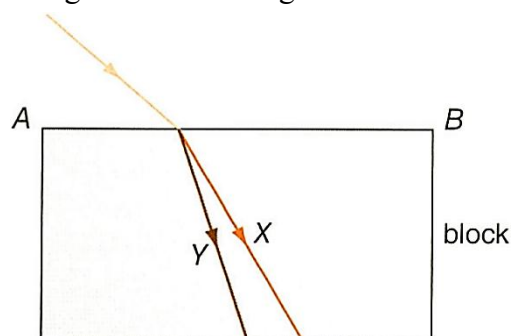
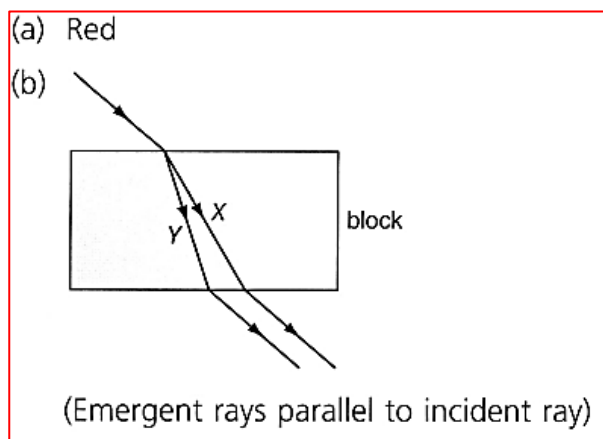
$\therefore$ (3) is correct.



B

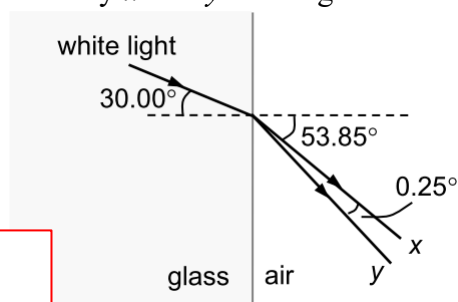
29. A light ray which consists of red light and violet light strikes a rectangular block and separate into two rays X and Y as shown. The refractive indices of the block for red light and violet light are 1.53 and 1.59 respectively.

- (a) What is the colour of X ?
(b) Draw the emergent rays of X and Y in the diagram.



30. A ray of white light travels from glass to air and a colour spectrum is formed as shown. Rays x and y are at the two ends of the spectrum. The angle between ray x and the normal is 53.85° and the angle between rays x and y is 0.25° .

- (a) What are the colours of rays x and y ? (2 marks)
(b) Do rays x and y travel at the same speed in air? (1 mark)
(c) The speed of white light in air is $2.998 \times 10^8 \text{ m s}^{-1}$. Find the speeds of ray x and y in the glass. The answers should be accurate to 4 significant figures. (4 marks)



- (a) x : red
 y : violet
(b) Yes
(c) By $n = \frac{c}{v}$ and $n_a \sin \theta_a = n_g \sin \theta_g$,

$$\sin \theta_a = \frac{c}{v} \sin \theta_g$$

$$v = \frac{\sin \theta_g}{\sin \theta_a} \times c$$

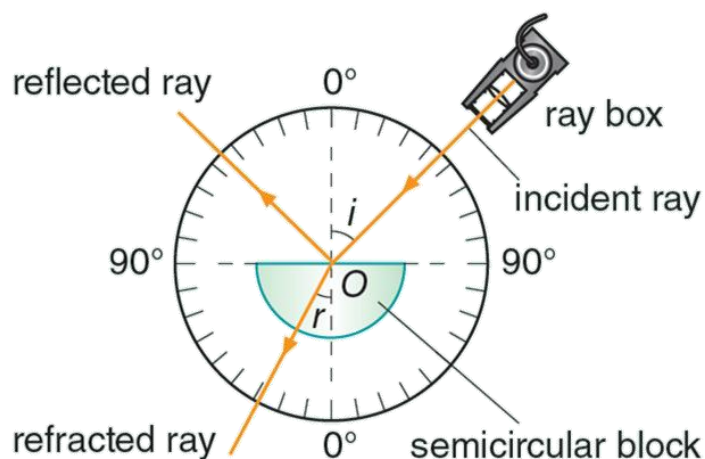
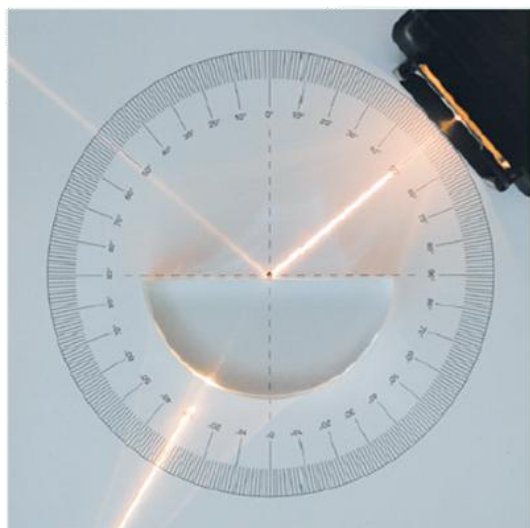
$$\text{Speed of ray } x = \frac{\sin 30.00^\circ}{\sin 53.85^\circ} \times 2.998 \times 10^8 = 1.856 \times 10^8 \text{ m s}^{-1}$$

$$\text{Speed of ray } y = \frac{\sin 30.00^\circ}{\sin (53.85^\circ + 0.25^\circ)} \times 2.998 \times 10^8 = 1.851 \times 10^8 \text{ m s}^{-1}$$

Experiment (Snell's law)

Snell's law forms a part of the laws of refraction: When a ray of light passes from one medium to another,

- the incident ray, the refracted ray and the normal all lie in the same plane.
- $\frac{\sin i}{\sin r} = \text{constant}$

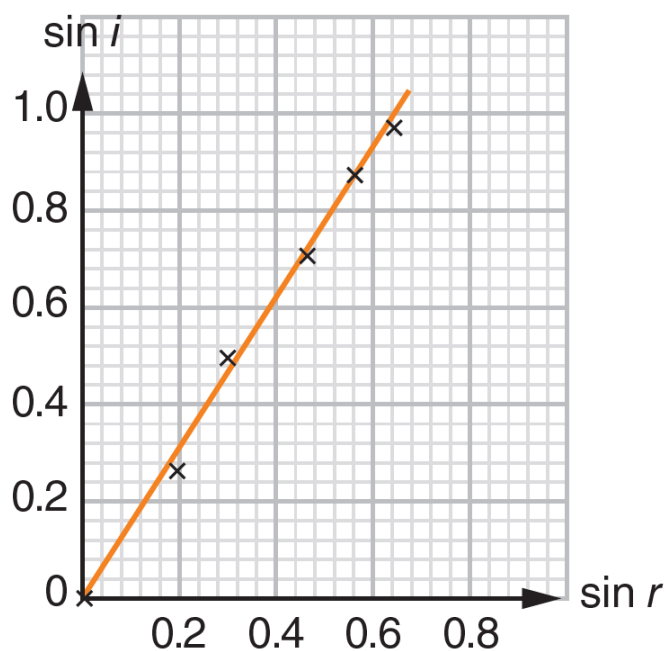


- ① Measure the angle of incidence i and angle of refraction r .
- ② Repeat step ① with other values of i .
- ③ Plot a graph of $\sin i$ against $\sin r$. find the value of $\frac{\sin i}{\sin r}$ for each pair of i and r .

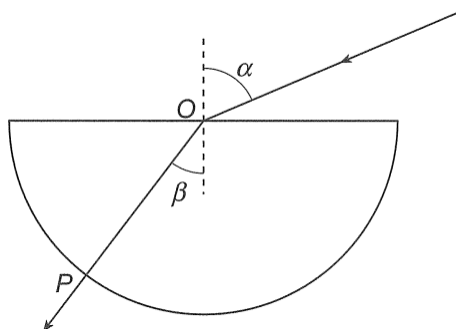
Remarks:

- Graph of $\sin i$ against $\sin r$:
straight line passing through the origin
- Snell's law:

$$\frac{\sin i}{\sin r} = \text{constant}$$



31.



A student performs an experiment to determine the refractive index of perspex. A ray of red light strikes a semi-circular Perspex block at the centre O. The angle of refraction is measured from the emergent ray. The values of α for different angles of incidence β are recorded in the following table.

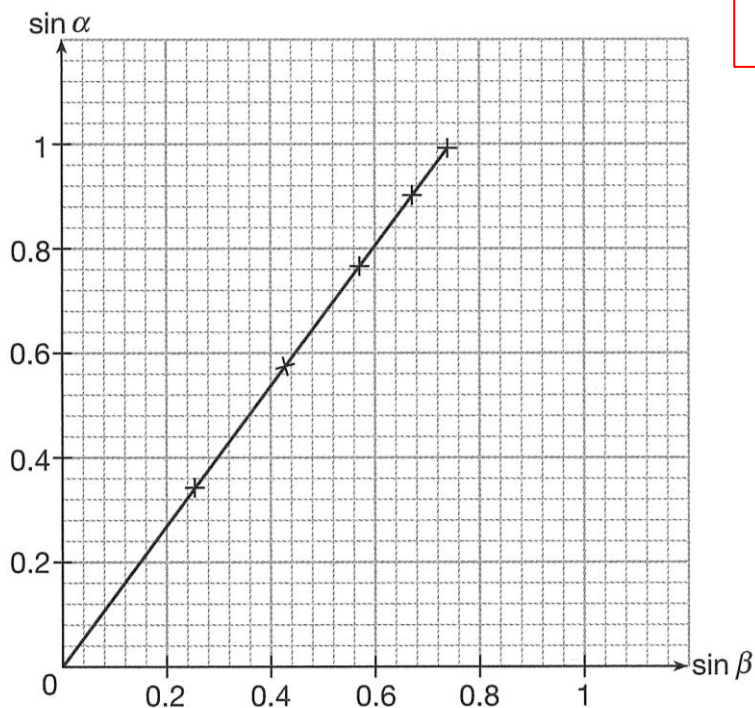
α	20°	35°	50°	65°	80°
β	14.7°	25.1°	34.6°	42.2°	46.8°
$\sin \alpha$					
$\sin \beta$					

- (a) Complete the table.
(b) Plot a graph of $\sin \alpha$ against $\sin \beta$.
(c) Determine the refractive index of perspex.

(a) Refer to the following table.

α	20°	35°	50°	65°	80°
β	14.7°	25.1°	34.6°	42.2°	46.8°
$\sin \alpha$	0.342	0.574	0.766	0.906	0.985
$\sin \beta$	0.254	0.424	0.568	0.672	0.729

(b) Refer to the following graph.

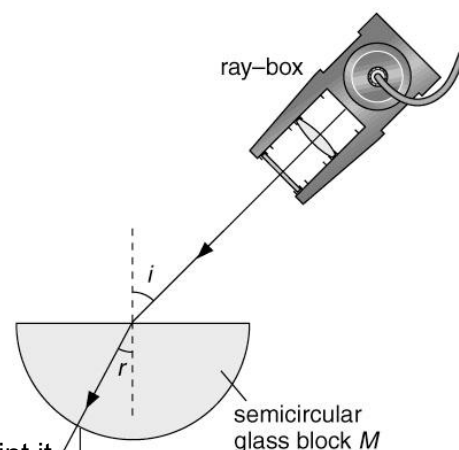


(c) Refractive index = $\frac{0.985}{0.729}$
= 1.35

32. The angles of refraction r are measured for different angles of incidence i . The results obtained are shown below:

i	20°	30°	40°	50°	60°
r	13°	18°	25°	32°	36°

- (a) Why is the light ray not bent when it leaves M ?
- (b) (i) Plot a graph of $\sin i$ (vertical axis) against $\sin r$ (horizontal axis) on a graph paper.
(ii) Find the slope of the graph.
(iii) State the physical meaning of the slope found in (ii).



- (a) This is because the light ray travels along the normal of the point it hits.

- (b) (i)

$\sin i$	0.34	0.50	0.64	0.77	0.87
$\sin r$	0.22	0.31	0.42	0.53	0.59

(ii)
$$\text{Slope} = \frac{0.84 - 0.3}{0.56 - 0.2}$$

$$= 1.5$$

- (iii) The slope found in (ii) is the refractive index of M .

